

GARRETT COMES

+1 (480) 845-9033 | garrett.comes@gatech.edu | garrettcomes24@gmail.com
[linkedin.com/in/garrett-comes](https://www.linkedin.com/in/garrett-comes) | github.com/GHawk1124 | garrettcomes.me

SUMMARY

I am an aerospace engineer trained at Georgia Tech with a propulsion concentration, equally at home in spacecraft fluid hardware and the software that analyzes it. I spent four summers at Lockheed Martin Space on flight programs including Dragonfly and Mars Sample Return. I build my own simulation tools in Rust, Python, and CUDA, from GPU CFD solvers to superconducting-coil multiphysics and orbital analysis, alongside independent research on thermodynamic computing hardware. My background is strong in engineering optimization and agentic AI tooling.

EDUCATION

Georgia Institute of Technology

Jul 2022 – Aug 2026

B.S. Aerospace Engineering, Concentration in Propulsion

Atlanta, GA

- Coursework: Jet & Rocket Propulsion, Computational Fluid Dynamics, Aerothermodynamics, Finite Element Analysis, Control System Design, Aircraft Flight Dynamics, Aerodynamics.

PROFESSIONAL EXPERIENCE

Lockheed Martin Space — Propulsion Engineering Intern, Civil & Commercial Space

Summers 2022 – 2025

Flight programs incl. Dragonfly and Mars Sample Return

Waterton Campus, Denver, CO

- Designed a piston-tank propellant gauging system in which hermetically sealed internal magnets are read by radiation-hardened Hall-effect sensors, owning magnetic circuit sizing, sensor selection, and the integration concept. Ran magnetic simulations driven by optimization methods to size the design against any given set of requirements, then correlated the analysis with bench testing.
- Ran propellant slosh and fluid-structure-interaction simulations of spacecraft tanks in Ansys, plus pure slosh CFD in FLOW-3D, and supported design of a slosh characterization test rig used to correlate model predictions against test data.
- Contributed to a green-propellant trade study evaluating hydrazine-replacement monopropellants across performance, materials compatibility, and handling and safety criteria.
- Designed and analyzed propulsion hardware and fluid-system components, including valves, interfaces, and test support equipment, with Creo CAD models, layouts, and drawings for propulsion assemblies.
- Supported propulsion hardware test campaigns, requirements verification, anomaly investigation, and cross-functional design reviews with systems, manufacturing, and test teams.

Boys & Girls Clubs of the Valley — Future Engineers Intern

Jul – Aug 2021

Onsite engineering and construction, Valley Metro Northwest light-rail expansion

Phoenix, AZ

PUBLICATIONS

- **SCRAP: Contactless Detumbling and Capture of Orbital Debris via HTS Coil** — extended abstract in preparation for AIAA SciTech 2027, 2026, with co-authors from Lockheed Martin, Astranis, and Boeing. [\[draft\]](#)

INDEPENDENT RESEARCH

Energy-Limited Library Coverage for Discrete Design | *Python, JAX, NumPy*

Jun 2026

- Authored a preprint measuring when thermodynamic sampling hardware (Extropic's TSU) beats GPUs on the energy cost of protein and drug design, reframing the comparison from per-sample accuracy to how much of a target design library a sampler covers per joule.
- Set a coverage requirement across 59 randomized $q=20$ Potts landscapes and showed the sampling stage costs about $10^6 \times$ less energy than a GPU for equal library coverage, then bounded the honest end-to-end ratio, which stays throttled by shared digital readout and matching rather than by sampling.
- Stress-tested on real RAS-superfamily coevolutionary physics (mean-field and pseudo-likelihood Potts models) and showed a localized-interface binder formulation restores full target separation and coverage. All results come from CPU simulation with review-gated claims, bit-for-bit reproduction, and 9/9 regression tests passing.

Masked-Diffusion LM $\times$ THRML Joint Decoding | *JAX, THRML, PyTorch*

Apr – May 2026

- Hybridized a pretrained masked-diffusion language model with block-Gibbs joint sampling over a factor graph on Extropic's THRML hardware simulator, where joint factor-graph decoding beat independent argmax by 7–15 pp on a held-out test split.
- Probed the “no classical escape” hypothesis and retired it honestly once warm-started AIS/SMC crossed every tested energy barrier, which redirected the work toward the coverage-per-energy result above.

ENGINEERING PROJECTS

- SCRAP — Orbital Debris Detumbling & Capture, Senior Design** | *Python, FEniCSx, Rust, Gmsh* **2025 – 2026**
– Developed a coupled electromagnetic–thermal quench simulation of the YBCO HTS pancake coil, using an H-formulation with E–J power-law resistivity, per-turn tape meshing, radiation and cold-finger BCs, quench detection, and adaptive time-stepping, to establish safe operating current margins.
– Trained DDPG reinforcement-learning agents for eddy-current detumbling of uncooperative tumbling targets.
- spacedb — Space-Object Database & Servicing Route Planner** | *Rust, egui, wgpu* **2025 – 2026**
– Desktop GUI and CLI over the full public space-object catalog from Space-Track, GCAT, and ESA DISCOS, with SGP4 and high-fidelity propagation, collision screening, decay prediction, debris value scoring, multi-target servicing-route optimization, and a wgpu-rendered interactive 3D Earth and orbit scene.
- Eddy-Current Braking System — Controls & Hardware** | *Rust no_std, Arduino Due, NGSolve* **2025**
– Built an eddy-current brake end to end with no_std Rust firmware on an Arduino Due, reading an AS5048A magnetic encoder over SPI and driving an H-bridge with PWM, with explicit control modes for manual duty, feedforward, and feedforward-plus-PI speed control, output clamps, slew limits, and target ramping.
– Paired the firmware with a live egui tuning dashboard over UART telemetry, plotting motion and internal controller signals in real time, plus Python/NGSolve magnetic simulations of the braking physics. The whole stack, from flashing to simulation, runs under Nix.
- Valurile — GPU Lattice-Boltzmann CFD Solver** | *Rust, CUDA* **2024 – present**
– Multi-crate LBM workspace with Esoteric Pull in-place streaming that roughly halves memory traffic per timestep, static mesh refinement, Smagorinsky–Lilly LES, SRT/TRT/MRT collision operators, airfoil and cylinder obstacles with drag and lift histories, and interactive egui/GPU visualization.
- fem.rust.ae — FEM Solver in Rust, live in the browser** | *Rust, WASM, Bevy* **2025**
– From-scratch structural FEM, a shell model of a cylinder under axial load, with sparse solves via faer, an interactive Bevy/egui 3D UI, mesh-refinement convergence studies, and a WebAssembly build running live in the browser.
- ai-cfd — GPU-Resident Flow Solver for ML Dataset Generation** | *Python, Triton* **2026**
– Staggered-grid incompressible cylinder-flow solver written entirely in Triton kernels, from momentum through projection to rendering, with adaptive timestep, Jacobi/CG/BiCGStab pressure options, per-step physics diagnostics, and train/val/test sweep generation for AI-CFD surrogate experiments.
- repx — eBPF Reproducible-Build Attestation** | *Rust, Aya eBPF* **2026**
– Kernel-level tracing of build process and file events canonicalized into Merkle-rooted attestations with content-addressed dataflow-DAG provenance. Verification reruns report divergent tree nodes rather than a bare pass or fail.
- any-nix-bootstrap — Trust-Minimized Software Bootstrap** | *POSIX sh, live-bootstrap, Guix, Nix* **2026**
– Bootstrap chain from an 813-byte auditable seed through GCC 15/musl/Guile, GNU Guix, and Nix to any nixpkgs package, fully offline from one hash-verified source manifest, with stage 1 executed end-to-end on real hardware against 314 hash-pinned distfiles.
- Stream Sim — Browser Fluid Simulation** | *JavaScript, Canvas, Pyodide* **2023**
– Grid-based incompressible-flow demo running live in the browser, with staggered velocity fields, pressure projection, semi-Lagrangian advection, airfoil obstacles, and streamline and smoke visualization under interactive controls.
- Nix Packaging for Claude Desktop & Claude Science** | *Nix flakes, NixOS* **2025 – 2026**
– Published [claude-desktop-nix](#) and [claude-science-nix](#), flakes packaging Anthropic's Linux apps for NixOS, incl. a bubblewrap overlay so the apps' own sandboxing resolves bash, the dynamic linker, and CA certificates.

SKILLS

Languages: Python, Rust, C++, MATLAB/Simulink, CUDA/Triton, Wolfram, LaTeX, shell
Simulation & CAE: Ansys Fluent and Mechanical, FEniCSx, NGSolve, Gmsh, SolidWorks, Creo, Fusion 360, Inventor
Scientific computing: JAX, PyTorch, NumPy/SciPy, wgpu/OpenGL, MPI/PETSc, probabilistic inference incl. Gibbs, AIS, and SMC, engineering optimization
Aerospace: space propulsion & fluid systems, valves, propellant management, orbital mechanics incl. SGP4 and high-fidelity propagation, CFD & slosh analysis, magnetic simulation & test
Systems & tooling: Linux/NixOS, Git, eBPF, embedded no_std Rust, agentic AI development with Claude Code, MCP servers, and custom agent frameworks

LEADERSHIP, CERTIFICATIONS & AWARDS

- Yellow Jacket Space Program** **2023 – 2024**
Propulsion — pressurant isolation valve design for a collegiate liquid-propellant space-shot vehicle *Georgia Tech*
- Apex Robotics, FRC and VEX** **2017 – 2022**
President & Lead Software Engineer — competition robots, outreach, fundraising, team coordination *Mesa, AZ*
- Stellar Xplorers** **2021 – 2022**
Team Captain — led team to one of ten national-finals spots in Houston *Mesa, AZ*
- Certifications:** SolidWorks CSWA · Autodesk Inventor · Autodesk Fusion 360 · Wolfram Language Level 1
Awards: CTE Presidential Scholar · National Semifinalist, 2022